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Introduction to Software Engineering (aka: OOP, OOD)

Ex1: Algorithm: Implementing a plane projection of 3D volume

Goal: Getting comfortable designing and implementing a not-so-well-defined algorithm in a robust and runtime-efficient manner.

Note: This assignment can be seen as an algorithmic challenge, yet here we are focusing on the design and the general quality of the code.

Problem statement: Given 3D raster data matrix (M), and a plane (p). Compute the 2D matrix (image) of the projection of p in V (note: this problem is NOT well-defined for implementation - a detailed design is needed).

Problem definition: Given a 3D matrix (M) of size sx*sy*sz byte values (the entry M[x][y][z] is an unsigned integer value of 8,16, 32 bits), where 1<=sx,sy,sz<2^10 and 0<=x<sx, 0<=y<sy, 0<=z<sz.

Implementation Requirements: Your implementation is secondary to the design; you can implement your code in any programming language over [Java, Python, C, C++]. Your code should be correct and stable. Given a matrix of size 500^3 bytes, a runtime of at least 100 fps is expected; +30 FPS is a **must** (a proper Java implementation leads to 300+ fps on an M2 Mac 2023).

Data: In this assignment, we will use NRRD format and data from the 3D Slicer open-source project. In particular, we encourage you to start with this NRRD (MRI) file.

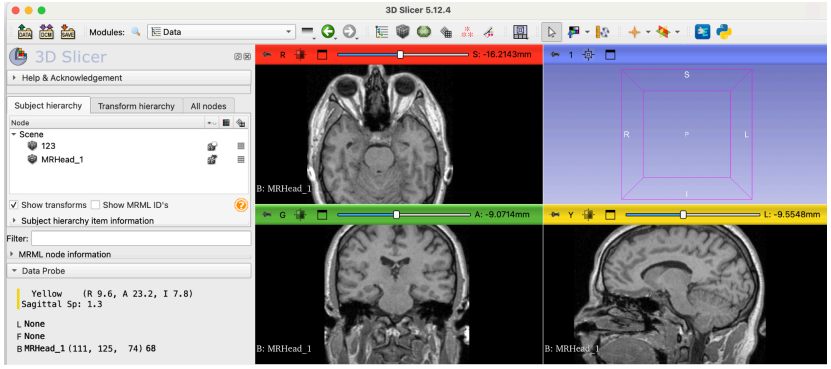


Figure 1: An example of a 3D NRRD file uploaded to the 3D Slicer software. The X, Y, Z Sections (Projections) are shown; your goal is to allow an efficient projection on any given plane.

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The Projection algorithm:

Given a 3D matrix (M), and a plane (p) the goal is to project M onto p as a 2D image (matrix).

Given a plane defined by 3 (3D) points (a,b,c) - where dist(a,b)=dist(a,c) (a,b,c is an isosceles triangle). Any point (q) within the triangle a,b,c that lies within M should be projected to the output image (2D matrix) using the following method:

1. Let Dab be the dist(a,b)=dist(a,c). Let Dbc be the dist(b,c).
2. Let v(a,b) be the normalized vector: vector(a,b) / Dab, define v(a,c) accordingly.
3. Define an output 2D Matrix E of size Dab*Dbc
4. Let a1=a, a2=a
5. For(int y=0; y<Dab; y++) # y runs from 0 (up) to Dab
    a. a1=a1+v(ab) # a1 steps from a to b
    b. a2=a2+v(ac) # a2 steps from a to c
    c. p=a1; # p steps from a1 to a2
    d. d12=dist(a1,a2) # d12 is the distance between a1 and a2
    e. dp=vector(a1,a2)/d12 # dp is the normalized vector(a1,a2)
    f. for(int x1=0;x1<=d12;x1++)
        i. v=M[p] # if p is NOT in M, v=0
        ii. x=(Dbc/2)-(d12/2) + x1 # x
        iii. E[x][y] = v
        iv. p=p+dp
6. Return E

Note: the same algorithm holds if instead of an isosceles triangle (a,b,c) the plane is presented by an isosceles trapezoid (two vectors on the same plane with the same length: a'→b and a''→c.)

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To Do:

1. Create a repository, define the problem and the scope of the project (all in the README).
2. Define an SRS document (Software Requirements Spec).
3. Define the main algorithm (in detail).
4. Define a data set of NRRD files.
5. Define a testing strategy; make sure to address the demo (as in 7)
6. Implement the project (Commit frequently).
7. Create a demo "main" allowing you to run the script in the following format: python Ex0.py file.nrrd file.csv file.mp4.
    a. The <file.nrrd> is a 3D volumetric data file in an NRRD format.
    b. The <file.csv> is a csv file representing a list of 3 or 4 points coordinates.
    c. The <file.mp4> is a video (output) file, including a frame for each corresponding plane coordinate as presented in the csv file (in the list order).
8. Perform an experiment and write down a short description of the results, including the mp4 output file.

Q & A

1. Can you elaborate on the definition of the plane (p)?

    Ans: P can be represented using 3 or 4 points (on the same plane:

    In case of 3 Points (a,b,c): |dist(a,b) - dist(a,c)| < EPS (EPS can be assumed to be 0.001).

    In case of four Points (a1, a2, b, c)

    a) |dist(a1,b) - dist(a2,c)| < EPS
    b) dist(a1,a2) <= dist(b,c)
    c) A1→b and a2→c are on the same plane (p)

2.